

Do Robots Have to be Human-Like? A Practice-Based Perspective on Relating to Robots In-the-Wild

Melissa M. Sexton
School of Business and Economics
Vrije Universiteit
Amsterdam, Netherlands
m.m.sexton@vu.nl

Anastasia V. Sergeeva
School of Business and Economics
Vrije Universiteit
Amsterdam, Netherlands
a.sergeeva@vu.nl

Maura Soekijad
School of Business and Economics
Vrije Universiteit
Amsterdam, Netherlands
m.soekijad@vu.nl



Figure 1: Food delivery practices, before and after robot introduction.

Abstract

Human-robot interaction has predominately applied a psychological lens to understanding how humans relate to robots, often referred to as anthropomorphism. However, this lens backgrounds the context-dependent and emerging ways humans relate to robots in real-world work settings. We draw on 17 qualitative interviews from an in-the-wild study of restaurant workers who interacted with autonomous mobile delivery robots for up to four years. Using practice theory, we shift focus from how workers perceive particular robot features towards examining the situated ways robotic features manifest as meaningful to workers for performing work. Our analysis highlights three dimensions of work that shape workers' relationship with robots: *temporality*, *physical space*, and *materiality*. We argue that a practice-based approach offers actionable insights for designing robots, whether anthropomorphic or otherwise, to integrate into holistic, situated work environments, look beyond isolated task-based use, and effectively support workers' needs.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720154>

CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; • **Computer systems organization** → Robotics.

Keywords

robots at work, in-the-wild, anthropomorphism, practice theory

ACM Reference Format:

Melissa M. Sexton, Anastasia V. Sergeeva, and Maura Soekijad. 2025. Do Robots Have to be Human-Like? A Practice-Based Perspective on Relating to Robots In-the-Wild. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3720154>

1 Introduction

The role and extent of anthropomorphic features for the design of robots intended to work alongside humans has been a topic of debate for many years amongst human-robot interaction (HRI) scholars. Anthropomorphism, broadly defined as “imbuing the imagined or real behavior of nonhuman agents with humanlike characteristics, motivations, intentions, and emotions” [13], has traditionally been understood through a psychological perspective. This perspective frames anthropomorphism as a cognitive process in which humans draw on prior experiences to relate to and interpret non-human objects [3]. Within robotics, such human-like features are often intended to promote trust [9, 19], enhance likability [5, 8, 23, 29],

enable more forgiveness for errors [10, 19, 30], or serve as storytelling tools to frame robots in familiar, human-like ways [10, 26]. In HRI studies, anthropomorphism has been typically measured by the humans' perception of the robot (i.e., likeability, intelligence, safety), attitudes towards the robot (i.e., trust, acceptance, empathy), by affect (i.e., activation, pleasure), or interactional behavior with the robot (i.e., task performance, social behavior) [23].

However, this lens of relating to robots often overlooks the situated and contextual dimensions of human-robot interaction in practice [6, 24]. Backgrounding contextual elements risks overlooking temporal aspects, such as how workers' interactions with robots evolve over time, instead focusing only on a snapshot, such as initial encounters with the robot [24]. Moreover, the situated work environment and the work being performed play a critical role, as not all robots or their features are equally suited to every work context. For instance, an experimental study of an industry robot revealed that anthropomorphic features did not always enhance trust and at times distracted the participants, raising safety concerns for real industrial use [21]. Similarly, in a medical setting, an experimental study revealed that patients were more embarrassed discussing their medical issues with an anthropomorphized robot compared to a non-humanlike technical box [4]. Findings like these underscore the context-sensitive nature of human-robot interaction, prompting questions about the necessity and appropriateness of the use of anthropomorphic features for robots, particularly within certain work environments.

To further unpack the in-situ components of the work setting in which the robots are used, a call remains open for more HRI studies to be conducted “in-the-wild,” especially over time [16]. Leaving the laboratory and experimental settings allows for better understanding of the contextual components missing from many previous HRI studies [6]. Studying a robot in practice can uncover unintended consequences, look beyond single human-robot interactions, and examine how the work changes based on how people interact with robots [16]. Thus, we present an in-the-wild empirical study of a restaurant that has used multiple autonomous mobile robots over the past four and a half years. To make sense of how workers relate to the robots, we turned to practice theory [20, 25]. This perspective allowed us to move beyond anthropomorphism and instead explore the situated ways robotic features manifested as meaningful to workers for performing work. Our analysis highlights three dimensions of work that shape workers' relationship with robots: *temporality*, *physical space*, and *materiality*, which we outline in the results.

1.1 Practice Theory

Practice theory emphasizes social life as constituted by organized, routinized activities that emerge and are sustained by practices, interconnected bundles of doings, sayings, and material arrangements. These activities do not occur independently but are combined as part of broader practices [20]. These practices are shaped by temporality—the past, present, and future—carrying traces of the past, unfolding in the present, and projecting toward the future. They are also inherently spatial, occurring within situated physical and social environments. Likewise, materiality—including an array of

material resources making the doing possible, such as objects, artifacts, technologies, but also human bodies—plays a central role, as material means are not passive tools but active participants in shaping and sustaining these practices [25].

Practice theory offers a powerful lens for exploring the entangled relationship between robots and everyday practices within in-the-wild work environments. It enables us to gain nuanced insights into how robots become embedded in the social and physical practices of work. By adopting this lens, we can trace the situated ways robotic features manifest over time by examining the variations across different contextual environments within a workplace and exploring how robots interact with other material objects. This shift of theoretical lens moves beyond examining anthropomorphic features solely, instead offering a holistic understanding of how technologies, like robots, are interwoven into the fabric of work.

2 Method

2.1 Setting and the Robots

Our study was conducted at the restaurant Sichuan Pagoda¹. This large restaurant spans three floors where four first-generation Pudubot delivery robots are used simultaneously (Figure 2). The owner has purchased a total of seven Pudubots since 2020 with the aim to lessen the physically demanding tasks for the workers, such as running back and forth from the kitchen to the main floor carrying multiple food dishes at once. Three robots operate on the ground floor, while one robot is used on the first floor. Each floor contains round and rectangular tables, arranged on the main lower floor and in an elevated area near the windows on the south, east and north sides of the restaurant accessible by a small set of stairs. Robots navigate exclusively on the non-elevated floor areas of the ground and first floor (Figure 2).

The Pudubot delivery robots are autonomous mobile robots that have a blend of human-like and machine-like features. Their non-humanoid bodies consist of four removable shelves (42x50 cm) and a 7-inch digital touchscreen for task input, such as assigning table numbers. The screens also display facial expressions both when idle or in motion. When unobstructed, the robots display either a “flat face,” a “smiling face,” or an animated “laughing face.” When robots encounter issues they display distressed faces, such as a “sad face,” a “crying face,” or an “angry face” (Figure 3). The robots also speak in a female voice, with selectable language and volume, to announce their movement, confirm their arrival at a destination, or even sing “Happy Birthday.”

The robots' movement further embodied a mix of human-like and machine-like qualities. The robots autonomously respond to obstructions using their sensors by either slowing down, rerouting, or stopping entirely. The robots also “communicate” their location with other robots via Wi-Fi to prevent collisions. The robots rely on pre-programmed paths guided by SLAM (Simultaneous Localization and Mapping) technology and ceiling-mounted sensors and have programmed “parking spots” for different purposes: unloading food at the floor zones, loading food at the pass², or waiting to go to the pass in “temporary parking” spots (Figure 2).

¹The names of the restaurant and all restaurant workers have been anonymized.

²The pass is a staging area where completed food orders are kept under heat lamps until the waitstaff picks them up for delivery to customers.

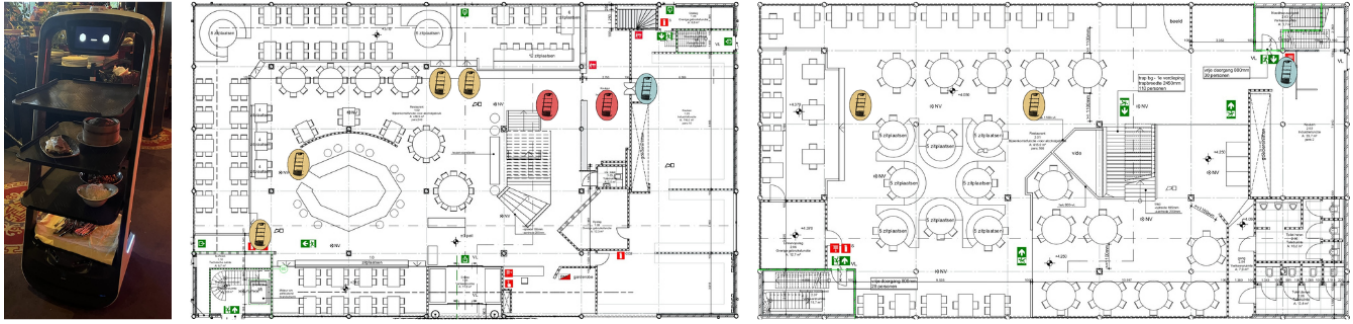


Figure 2: Image of a Pudubot and an image of the floor maps of the restaurant's ground floor and first floor. The gold robots represent robots' food unloading spots (floor zones), the blue robot represents the loading spot (the pass), and the red robots represent robots waiting to go to the pass ("temporary parking").

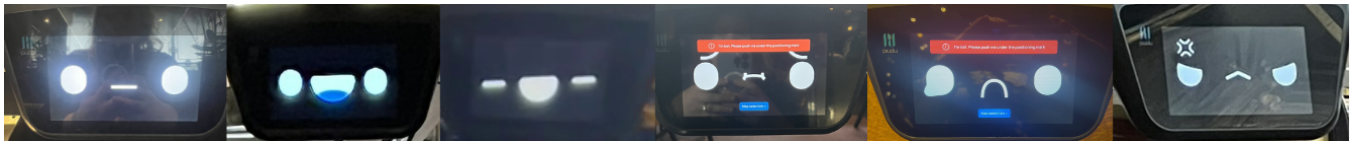


Figure 3: Image of various Pudubot facial expressions. From left to right, "flat face," "smiling face," "laughing face," "sad face," "crying face," and "angry face."

2.2 The Practice of Delivering Food

The robots replaced an existing physical, manual practice with a digital, semi-automatic practice (Figure 1). Before robots, foodrunners loaded food dishes with their respective order tickets onto a wheeled cart at the pass, then manually pushed the cart to the restaurant floor. Once on the floor, foodrunners and waiters unloaded the cart and delivered the food to the respective customers. Now with robots, the responsibility of loading food falls to a single pass employee, who arranges the dishes onto the robot's trays inside the kitchen. To ensure maximum efficiency, the pass employee arranges the dishes in a "Tetris-like" manner, as each robot shelf is designed to accommodate only one table. The pass employee enters the table number on the robot's digital screen and confirms the order. The robot autonomously travels across the restaurant floor transporting the food to one of the unloading spots (Figure 2). Although the robot does not move directly between tables, obstacles such as customers shifting chairs or hanging coats can narrow the available space on the restaurant floor. Upon the robot's arrival to an unloading spot, a foodrunner or waiter unloads the food, confirms the task's completion on the digital screen, and delivers the food to the customer³. The robot then autonomously proceeds to its next designated zone or, if empty, returns to a loading spot.

2.3 Interviews and Analysis

The first author conducted 17 qualitative semi-structured interviews with restaurant workers at Sichuan Pagoda who had interacted with the robot for up to 4.5 years ($M = 2.90$ years, $SD = 1.81$ years). Participants included various staff members as well as the restaurant

owner to gain insight into managerial decisions regarding the adoption and use of the robots (Appendix A.1). Interviews were recorded after receiving consent and averaged 43 minutes. The interview questions first focused on the participants' daily work practices, e.g., delivering food, followed by targeted questions about their perceptions of the robots, examining how these perceptions evolved from the initial interaction with the robots to the present.

We began by analyzing the interview data by comparing workers' perceptions of the robots as either machine-like or human-like [12] using the broad anthropomorphic categories of physical appearance, emotion, voice, movement, touch, and intent. As the analysis progressed, it became evident that intent and emotion were not distinct categories but instead interwoven within other anthropomorphic features, such as face, voice, and movement. As we analyzed the data, we found that workers rarely focused on specific features of the robots. Instead, their perceptions centered on their experiences of work and how their relationships with the robots evolved over time. To systematically categorize these experiences, we identified instances where the robots either augmented or constrained workers' ability to perform food delivery, leading us to construct Table 1.

Next, examining all instances where the robots augmented or constrained work, we inductively coded [28] employees' explanations of when and why the robots' features, whether machine-like or human-like, augmented and constrained their work, such as the fast-pace tempo of the restaurant or physical obstructions on the restaurant floor. Inductively aggregating these conditions, we identified three broader dimensions of work that shaped workers' relationships with the robots in practice: *temporality*, *physical space*, and *materiality* (Table 2).

³Customers do not remove food from the robots, only restaurant workers.

Table 1: Restaurant Workers' Perceptions of Robotic Features: Augmenting and Constraining Work

		Robot Body	Robot Face (as emotion)	Robot Voice (as emotion)	Robot Voice (as intent)	Robot Movement (as intent)
Machine-like	Affords...	Carrying more plates Carrying hot dishes Not getting tired Clear communication of table numbers Being organized and compact A physical presence	<i>Does not manifest as meaningful for the practice of delivering food</i>	One set volume Repetitious speaking Programmed phrases	One set volume Repetitious speaking Programmed phrases	Programmed routes Programmed slow speeds (for safety protocols) Non-adjustable speeds
	Augments work by...	Delivering food quicker to customer Workers walking less Workers not burning hands Workers not having to pick up slack from a 'colleague' Providing easy readability of table numbers Workers not getting sticky hands from order tickets Making table orders easier to remember Providing a visual reminder for food unloading	<i>Does not manifest as meaningful for the practice of delivering food</i>	Not arguing, just does job	<i>Does not manifest as meaningful for the practice of delivering food</i>	Providing predictability to anticipate movement
	Constrains work by..	Becoming an obstacle to squeeze past in constrained spaces Workers needing to bend down for bottom shelves	<i>Does not manifest as meaningful for the practice of delivering food</i>	<i>Does not manifest as meaningful for the practice of delivering food</i>	Becoming drowned out and ignored Not being applicable to the situated work	Delivering food slower to customers due to slow movement speed Not adjusting to the situated environment (e.g., busyness)
Human-like	Affords..	<i>Does not manifest as meaningful for the practice of delivering food</i>	Conveying 'emotions'	Expressing politeness	Expressing intent to complete tasks Asking for help	Autonomous choices of movement Stopping/slowng for obstacles
	Augments work by...	<i>Does not manifest as meaningful for the practice of delivering food</i>	Bringing humor 'Brightening' work	Providing positivity for less work stress Attracting guests who become visual reminders for food unloading	<i>Does not manifest as meaningful for the practice of delivering food</i>	Workers trusting work by robots to be done autonomously
	Constrains work by...	<i>Does not manifest as meaningful for the practice of delivering food</i>	Attracting guests who become obstacles	<i>Does not manifest as meaningful for the practice of delivering food</i>	Becoming drowned out and ignored	Delivering food slower to customers after becoming stuck from unpredictable movements Bumping into humans Workers having to wait for the robot to move Spilling liquid items (e.g., soup, drinks)

3 Results

3.1 Initial Perceptions

Although the robots were purchased for their utility rather than their novelty, the “novelty effect” initially shaped the perceptions of many workers, who were surprised and excited to see the robots

in action at the restaurant. Several workers described the robots as “*pretty cool technology*” and reacted most strongly to the robots’ facial expressions, which they either found “*cute*,” “*funny*,” or, in some cases, “*creepy*.” However, as seen in previous studies, these novelty reactions quickly faded over time [11, 18, 27] and workers

began to relate to the robots through the lens of their work practices, specifically the practice of delivering food.

These various work-related manifestations were shaped by temporal elements, aspects of the situated physical space, and the materiality of the work. While we categorically separate the categories of temporality of work, physical space of work, and materiality of work in our results, it is important to note that many of the examples were intertwined with one or all the categories.

3.2 Temporality of Work

Restaurant workers perceived and interacted with the robots differently depending on the time of the day, day of the week, or duration of their shift. For instance, Eve explained that on Sundays, when customers often ordered one dish at a time, robots were seen as less efficient, as they required multiple trips to serve large tables. Additionally, workers' perceptions of anthropomorphic features changed throughout their shifts. Robots' "cute" faces could provide a welcome mood boost during long hours, as Mara noted, *"If you're tired, you've been working for seven hours and then you see a robot, like smiling, it's kind of fun."* However, over the course of a long shift, the robots' voices could become irritating, Mara added, *"If you're working ten hours and you hear that [the robot's voice] all the time, it is kind of annoying."*

The tempo of restaurant work further shaped workers' interactions with the robots. During peak hours, the robots became "invisible" to workers amidst the hustle and bustle. Kristina observed, *"I think when it's busier, people [workers and customers] will not notice it [the robot] (...) I think when it's less busy, it's [the robot] visible."* Calvin reiterated the notion of invisibility describing robots as "food coming" and as part of "the whole system working with us." He explained, *"They're [the robots] just there. Sometimes I forget that there are actually robots doing it [the food delivery]."* However, during slower times, the robots became a helpful signal for "ready-to-deliver food." David shared in slower times the robot's presence helped him refocus on his primary job of delivering food when engaged in a secondary task. He explained, *"When it's not that busy, in the middle [the restaurant floor], I have to do my carts [dirty dishes] and go back [to dishwashing] and oh, I see my food that I have to run already on the robot, nice, nice."*

The speed of work in the restaurant also played a role in influencing how workers related to the robots. Some workers criticized the robots' slow pace and frequent stopping as disruptive to the workflow such as Geena who likened the robots to a "slow cart" that could cause spills due to its sudden stops. However, others appreciated the robots for speeding up food delivery, especially during busy times. Carlos, emphasized the robots' ability to expedite food delivery and prevent dishes from sitting idle at the pass, *"It definitely speeds up the process (...) you can have dishes that stand in the kitchen for too long and with these robots, they can always be sent down to the restaurant."* Mara further noted that the robots helped distribute workload across the restaurant, creating a smoother workflow expressing *"the busyness is leveled in different places."*

⁴Designer recommendations may sometimes conflict across different dimensions of work, requiring careful consideration of which changes to implement based on their overall impact to the workflow.

3.3 Physical Space of Work

The physical space of work was a factor in how workers related to the robots. Distance played a large role, as the robots were introduced to help mitigate the long walking distance between the kitchen and the restaurant floor. Joseph explained how he now depends on the robots to reduce his walking distance at work, *"So you can imagine, like without the robots, at least 30-40% extra walking distance."*

Routes, both those of the workers and the robots, were another key factor influencing how workers related to the robots. Workers valued the predictability of the robots' machine-like programmed paths, which enabled them to anticipate and adapt to their movements while unpredictable robot movements were undesirable. Gilbert explained, *"Everybody knows where he [the robot] stops, how he turns right, left (...) If he stops anywhere [unpredictable], that's also not good."* Over time, workers developed strategies to harmonize their movements with those of the robots. Joseph shared how over time he learned to navigate more efficiently around the robots, *"At first, you're going to anticipate where the robots are going, and the robots are going to anticipate as well. (...) So, I learned like, just walk straight, it'll [the robot] move out of the way."* However, disruptions still occurred, and workers sometimes viewed the robots as obstacles when their paths intersected, whether in predictable or unpredictable routes. Carlos expressed appreciation for the robots but noted frustrations when they *"intervene with your path."*

The location of the robots within the restaurant was also influential in the relationship to the robots. Whether on the first level, ground level, at the pass, or on the general restaurant floor, robots elicited varying reactions depending on their specific location. For example, Joseph described the robots downstairs as critical to the work, while upstairs they were nice but not necessary. He explained, *"Downstairs you're like dependent on it [the robots], upstairs it's more like a helping hand basically."* However, certain areas, such as the tight opening between prominent tables, were more prone to disruption. David shared, *"Sometimes it's busy, so I have to move fast but the robots are sometimes in the way. Especially between table 16 and 19."*

3.4 Materiality of Work

Material objects of work influenced how workers related to the robots. Food was the primary material object associated with the robots, to the extent that workers often used the term "the food" interchangeably to refer to the robot itself. Rocky explained, *"The purpose of the robots, bringing the food to the table, actually, that's the most important thing."* Carlos also highlighted the advantage of robots being more machine-like in terms of carrying food, *"It's definitely nice because it can take multiple dishes at once. If you're a good waiter you can take five plates, but that's really good if you can do that. If you're normal good, you take four, and average is three plates. And you know, the robot can take many more [than five plates]."* Furthermore, when the robot failed to return to the pass because it was stuck on the floor due to not being unloaded or because of a malfunction, it caused problems in the kitchen and interfered with the flow of food delivery. Mindy shared, *"If the robot comes back [to the pass], I can work more. If [it does] not come back, (...) I want to cry, I say, 'where's my robot?'"*

Table 2: Practice-Based Work to Robotic Designer Suggestions

	Practice-Based Work	Robot Features	Work is constrained by...	Designer Recommendations ⁴
Temporality	Clock time	Body and face	... human-like faces attracting customers, disrupting the workflow during busy restaurant hours.	<i>Less attention-grabbing features to avoid unintended consequences such as customers getting in the workers' way as they interact with the robot during busy times.</i>
	Tempo of restaurant	Voice	... human-like voice becoming drowned out by restaurant busyness. ... human-like voice becoming annoying after working many hours.	<i>Alternatives to voice such as textual or visual communication of information [7] and/or expressive lighting [2] to call attention to unload robot during busy times.</i>
	Tempo of work	Movement	... machine-like movement being too slow during busy restaurant hours. ... human-like reactions to obstacles stopping the robot for too long disrupting the workflow during busy restaurant hours.	<i>Adaptable movement to the temporality of the situated work to move faster or slower depending on restaurant busyness or obstacles.</i>
Physical Space	Distance	Body and face	... machine-like body size becoming an obstacle by blocking tight entryways.	<i>Slimmer machine-like body that accommodates food but can still maneuver through tight spaces.</i>
	Routes	Voice	... not existing, workers desire human-like voice as a signal for coming up behind.	<i>A less repetitive voice that adapts to the contextual environment to adjust volume and tone [22] such as a warning for coming up behind an employee.</i>
	Location	Movement	... human-like autonomous movement causing robot to become stuck in constrained spaces. ... human-like autonomous movement creating paths that are unpredictable causing people to bump into the robot.	<i>Identify "choke points" in the environment and avoid programmed routes for the robot where it may become stuck. Maintain robotic predictable paths so workers can predict robotic movements that allow robots to become part of the process or flow of work.</i>
Materiality	"The food"	Body and face	... machine-like programmed phrases becoming ignored because it does not match the intention of the objects on the robot.	<i>Better adaptability of phrases to match the situated work [17] and materials that the robot is carrying, such as saying 'the soup is on tray two, be careful it is hot.'</i>
	Workers' bodies	Movement	... human-like autonomous movement causing liquids such as soup or drinks to spill.	<i>More human-like adaptable movement to move faster or slower depending on the material objects the robot is carrying [14], such as soups or drinks.</i>

Workers also related to the robots through other material obstructions, objects like chairs, coats, or people moving about (customers, children, or colleagues) that interfered with the robots' paths. Geena described some of these challenges, *"If people are sitting over there and they have [a] coat hanging over there and then the robot is stuck. Or maybe the children are very curious [and playing with the robot]. So it would be stuck over there."*

Additionally, the materiality of the workers' bodies frequently emerged as a key factor in how workers related to the robots. Workers viewed the robots as ways to alleviate physical strain, reducing excessive walking, easing the burden of heavy loads, preventing burns from hot dishes, or avoiding sticky order tickets. Emily described the physical relief provided by the robots, *"Every time that I work, I have like 20,000 steps from like one workday only. And I think without the robots it would be like 30,000 or something."* She also noted how the robots can carry more food, especially hot dishes, *"Sometimes the food is also really hot, so then I can't even take three dishes because then if you put, you like, really burn yourself."* Additionally, Alan described that with the robots' digital screens, floor staff avoided *"sticky hands"* as they did not have to pick up order tickets that may have touched the food.

4 Discussion

In response to the prevailing focus on the human-like features of robots, this in-the-wild study specifically examined how anthropomorphic design influences human-robot interaction in a work setting. Our analysis offers a new perspective on how workers engage with human or non-human features of robots in the course of their daily work drawing on the experiences of workers who have interacted with robots for up to four and a half years. From these experiences, we highlight how human-robot interaction is deeply shaped by the physical space, tempo, speed, and materiality of work practices. While this approach does not dismiss the potential value of anthropomorphic design features, it does caution against incorporating such features without a contextual understanding of the work itself (Table 2).

A practice-based perspective on human-robot interaction offers valuable guidance for designers, helping them identify which anthropomorphic features enhance work in certain environments or when they may become detrimental to the work. For example, human facial features on a robot might capture visual attention [1] which can be advantageous for engaging customers. However, in a restaurant setting, such distractions may disrupt work, negatively

affecting the overall workflow of multiple restaurant workers. Additionally, this practice-based approach can help identify which human-like features may prove ineffective in specific environments and whether machine-like or other creative options might be more suitable. For example, in noisy, fast-paced restaurant settings, voice features may be ineffective. Instead, expressive lighting [2] could offer a more effective alternative for robotic communication.

Taking a practice-based approach also extends beyond the simple selection of anthropomorphic features tailored to a single task [15]. Practices are embedded in a broader, interconnected network of actions, rather than being confined to isolated tasks. By examining the various dimensions of practices—temporality, physical space, and materiality—designers and users can make more informed decisions about what robotic features best align with the complexities of a dynamic work environment as a whole, rather than focusing solely on individual tasks.

Moreover, by emphasizing work practices, this approach expands the focus to explore the unintended consequences of robotic integration. It allows designers to gain a nuanced understanding of how robots interact with workers' established and emergent work practices. This shift moves the emphasis from simply delivering food quickly, for instance, to developing robot features that genuinely support workers' needs, enhancing the quality and sustainability of their work rather than merely accelerating it.

Acknowledgments

We express our gratitude to all the participants from Sichuan Pagoda for their time, sincerity, and invaluable contributions to our research. This research received funding from the NWO, grant no. VI.Vidi.211.255.

References

- [1] Jae-eul Bae and Myung-sun Kim. 2011. Selective visual attention occurred in change detection derived by animacy of robot's appearance. In *Proceedings of the 2011 International Conference on Collaboration Technologies and Systems (CTS '11)*, May 23–27, 2011, Philadelphia, PA, USA. IEEE, New York, NY, USA, 190–193. <http://doi.org/10.1109/CTS.2011.5928686>
- [2] Kim Baraka and Manuela M. Veloso. 2018. Mobile service robot state revealing through expressive lights: Formalism, design, and evaluation. *Int. J. Soc. Robot.* 10 (Jan. 2018), 65–92. <http://doi.org/10.1007/s12369-017-0431-x>
- [3] Christoph Bartneck, Tony Belpaeme, Friederike Eyssel, Takayuki Kanda, Merel Keijsers, and Selma Sabanovic. 2024. *Human-Robot Interaction: An Introduction* (2nd. ed.). Cambridge University Press, Cambridge, UK. <https://www.human-robot-interaction.org/>
- [4] Christoph Bartneck, Timo Bleeker, Jeroen Bun, Pepijn Fens, and Lynnyrd Riet. 2010. The influence of robot anthropomorphism on the feelings of embarrassment when interacting with robots. *Paladyn J. Behav. Robotics* 1 (Aug 2010), 109–115. <http://doi.org/10.2478/s12320-010-0011-3>
- [5] Christoph Bartneck, Dana Kulic, Elizabeth Croft, and Susana Zoghbi. 2009. Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *Int. J. Soc. Robot.* 1 (Jan 2009), 71–81. <http://doi.org/10.1007/s12369-008-0001-3>
- [6] Samia Cornelius Bhatti and Lionel Peter Robert. 2023. What does it mean to anthropomorphize robots? Food for thought for HRI research. In *Proceedings of Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI'23)*, March 13–16, 2023, Stockholm, Sweden. ACM, New York, NY, USA, 422–425. <https://doi.org/10.1145/3568294.3580119>
- [7] Andrea Bonarini. 2020. Communication in human-robot interaction. *Curr Robot Rep* 1 (Dec 2020), 279–285. <http://doi.org/10.1007/s43154-020-00026-1>
- [8] Alvaro Castro-Gonzalez, Henny Admoni, and Brian Scassellati. 2016. Effects of form and motion on judgments of social robots animacy, likability, trustworthiness and unpleasantness. *Int. J. Hum. Comput. Stud.* 90 (June 2016), 27–38. <http://doi.org/10.1016/j.ijhcs.2016.02.004>
- [9] Lara Christoforakos, Alessio Gallucci, Tinatini Surmava-Große, Daniel Ullrich, and Sarah Diefenbach. 2021. Can robots earn our trust the same way humans do? A systematic exploration of competence, warmth, and anthropomorphism as determinants of trust development in HRI. *Front. Robot. AI* 8, Article 640444 (April 2021), 15 pages. <https://doi.org/10.3389/frobot.2021.640444>
- [10] Kate Darling. 2017. "Who's Johnny?" Anthropomorphic framing in human-robot interaction, integration, and policy. In *Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence*, Patrick Lin, Keith Abney, and Ryan Jenkins (Eds.). Oxford University Press, New York, NY, USA, 173–188. <https://doi.org/10.1093/os0/9780190652951.003.0012>
- [11] Maartje MA de Graaf, Somaya Ben Allouch, and Jan AGM van Dijk. 2018. A phased framework for long-term user acceptance of interactive technology in domestic environments. *New Media Soc.* 20 (July 2018), 2582–2603. <http://doi.org/10.1177/1461444817727264>
- [12] Judith Dörrenbächer, Diana Löffler, and Marc Hassenzahl. 2020. Becoming a robot - overcoming anthropomorphism with techno-mimesis. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*, April 25–20, 2020, Honolulu, HI, USA. ACM, New York, NY, USA, 1–12. <http://doi.org/10.1145/3313831.3376507>
- [13] Nicholas Epley, Adam Waytz, and John T. Cacioppo. 2007. On seeing human: A three-factor theory of anthropomorphism. *Psychol Rev.* 114 (Oct 2007), 864–886. <http://doi.org/10.1037/0033-295x.114.4.864>
- [14] Luca Garello, Linda Lastrico, Francesco Rea, Fulvio Mastrogiovanni, Nicoletta Noceti, and Alessandra Sciutti. 2021. Property-aware robot object manipulation: A generative approach. In *Proceedings of the 2021 IEEE International Conference on Development and Learning (ICDL '21)*, Aug. 23–26, 2021, Beijing, China. IEE, New York, NY, USA, 55–60. <http://doi.org/10.1109/ICDL49984.2021.9515667>
- [15] Jennifer Goetz, Sara Kiesler, and Aaron Powers. 2020. Matching robot appearance and behavior to tasks to improve human-robot cooperation. In *Proceedings of the 12th IEEE International Workshop on Robot and Human Interactive Communication (RO-MAN '03)*, Oct. 31–Nov. 2, 2003, Millbrae, CA, USA. IEE, New York, NY, USA, 55–60. <http://doi.org/10.1109/ROMAN.2003.1251796>
- [16] Malte Jung and Pamela Hinds. 2018. Robots in the wild: A time for more robust theories of human-robot interaction. *ACM Trans. Hum.-Robot Interact.* 7 (May 2018), 5 pages. <https://doi.org/10.1145/3208975>
- [17] Linda Lastrico, Luca Garello, Francesco Rea, Nicoletta Noceti, Fulvio Mastrogiovanni, Alessandra Sciutti, and Alessandro Carfi. 2021. Robots with different embodiments can express and influence carefulness in object manipulation. In *Proceedings of the 2022 IEEE International Conference on Development and Learning (ICDL '22)*, Sept. 12–15, 2022, London, UK. IEE, New York, NY, USA, 55–60. <https://ieeexplore.ieee.org/document/9962196>
- [18] Iolanda Leite, Carlos Martinho, Andre Pereira, and Ana Paiva. 2020. As time goes by: Long-term evaluation of social presence in robotic companions. In *Proceedings of the 18th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN '09)*, Sept. 27–Oct. 2, 2009, Toyama, Japan. IEE, New York, NY, USA, 669–674. <http://doi.org/10.1109/roman.2009.5326256>
- [19] Manisha Natarajan and Matthew Gombolay. 2020. Effects of anthropomorphism and accountability on trust in human robot interaction. In *Proceedings of the 4th ACM/IEEE International Conference on Human-Robot Interaction (HRI '20)*, March 23–26, 2020, Cambridge, UK. ACM, New York, NY, USA, 33–42. <https://doi.org/10.1145/3319502.3374839>
- [20] Davide Nicolini. 2012. *Practice Theory, Work, and Organization: An Introduction*. Oxford University Press, Oxford, UK.
- [21] Linda Onnasch and Clara Laudine Hildebrandt. 2021. Impact of anthropomorphic robot design on trust and attention in industrial human-robot interaction. *ACM Trans. Hum.-Robot Interact.* 11, Article 2 (Oct. 2021), 24 pages. <https://doi.org/10.1145/3472224>
- [22] Qiaoqiao Ren, Yuanbo Hou, Dick Botteldooren, and Tony Belpaeme. 2024. No more mumbles: Enhancing robot intelligibility through speech adaptation. *IEEE Robot. Autom. Lett.* 9 (July 2024), 6162–6169. <http://doi.org/10.1109/LRA.2024.3401117>
- [23] Eileen Roesler, Dietrich Manzey, and Linda Onnasch. 2021. A meta-analysis on the effectiveness of anthropomorphism in human-robot interaction. *Sci. Robot.* 6 (Sept. 2021), 10 pages. <http://doi.org/10.1126/scirobotics.abj5425>
- [24] Eileen Roesler, Lara Naendrup-Poell, Dietrich Manzey, and Linda Onnasch. 2022. Why context matters: The influence of application domain on preferred degree of anthropomorphism and gender attribution in human-robot interaction. *Int. J. Soc. Robot.* 14 (Jan. 2022), 1155–1166. <http://doi.org/10.1007/s12369-021-00860-z>
- [25] Theodore R. Schatzki. 2012. A primer on practices: Theory and research. In *Practice-Based Education: Perspectives and Strategies*, Ronald Barnett, Stephen Billett, Maggie Hutchings, and Franziska Trede (Eds.). Sense Publishers, Rotterdam, NL, 13–26. https://link.springer.com/chapter/10.1007/978-94-6209-128-3_2
- [26] Sarah Schömb, Jacobe Klein, and Eileen Roesler. 2023. Feeling with a robot—the role of anthropomorphism by design and the tendency to anthropomorphize in human-robot interaction. *Front. Robot. AI* 10 (June 2023), 16 pages. <http://doi.org/10.3389/frobot.2023.1149601>
- [27] JaYoung Sung, Henrik I. Christensen, and Rebecca E. Grinter. 2009. Robots in the wild: Understanding long-term use. In *Proceedings of the 4th ACM/IEEE International Conference on Human-Robot Interaction (HRI '09)*, March 11–13, 2009, La Jolla, CA, USA. ACM, New York, NY, USA, 45–52. <http://doi.org/10.1145/>

- 1514095.1514106
- [28] Sarah J. Tracy. 2019. *Qualitative Research Methods: Collecting Evidence, Crafting Analysis, Communicating Impact* (2nd. ed.). Wiley-Blackwell, Chichester, UK.
- [29] Jinchun Wu, Xiaoxi Du, Yixuan Liu, Wenzhe Tang, and Chengqi Xue. 2024. How the degree of anthropomorphism of human-like robots affects users' perceptual and emotional processing: Evidence from an EEG study. *Sensors* 24 (July 2024), 26 pages. <http://doi.org/10.3390/s24154809>
- [30] Kai Chi Yam, Yochanan E Bigman, Pok Man Tang, Remus Ilies, David De Cremer, Harold Soh, and Kurt Gray. 2021. Robots at work: People prefer—and forgive—service robots with perceived feelings. *J. Appl. Psychol.* 106 (Oct. 2021), 1557–1572. <http://doi.org/10.1037/apl0000834>

5 Appendices

5.1 Interview Table

Table 3: Restaurant Workers' Roles and Experience

Name	Role(s)	Employment	Time Working with Robots	Prior Experience with the Robots	Interview Length (min)
Alan	Manager	15 years	4.5 years	No	70
Bianca	Host	4.5 years	4.5 years	No	56
Calvin	Waiter	1 year	1 year	No	46
Carlos	Bartender	1 month	1 month	No	36
David	Foodrunner	4 months	4 months	No	26
Emily	Foodrunner	2 years	2 years	No	34
Eve	Supervisor, waiter, host	5 years	4.5 years	No	40
Geena	Foodrunner, waiter	3 years	3 years	No	47
Gilbert	Waiter	24 years	4.5 years	No	36
Joseph	Waiter	6 months	6 months	Yes	43
Ken	Owner	-	-	No	71
Kristina	Foodrunner, waiter	6 years	4.5 years	No	41
Mara	Foodrunner	1 year	1 year	No	58
Mindy	Pass, waiter	11 years	4.5 years	No	25
Rocky	Pass, waiter	6 years	4.5 years	No	41
Teddy	Bartender	2.5 years	2.5 years	No	51
Tyler	Manager	11 years	4.5 years	No	43